

Metrology 4.0 and its central role in Industry 4.0

A brief review

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Abstract

Metrology 4.0 embeds metrological methods into smart manufacturing environments, using connected devices, automated workflows, and streaming data analytics to inform production decisions. Its enabling technologies include networked sensors, machine learning and AI, digital twin models, and cloud-native services that underpin measurement workflows. The transition to digital metrology introduces new benefits—including real-time quality control, improved productivity, collaborative innovation, and reduced costs—while also presenting challenges related to data security, interoperability, and model transparency. Digital metrology's maturation supports smart factory operations and could play an important role in the transition to Industry 5.0, which prioritizes collaboration and trust.

Keywords: Metrology 4.0, Industry 4.0.

1 Introduction

Metrology 4.0 refers to the application of metrological principles within the Fourth Industrial Revolution (Industry 4.0), where digital technologies reshape manufacturing practices. The evolution of metrology from the third to the fourth industrial revolution is a profound paradigm shift, characterized by digitalization, the integration of measurement systems into real-time production processes, and the intensive use of advanced technologies (Wieczorowski & Trojanowska, 2021). This evolution is based on the need to maintain maximum accuracy and traceability of the International System (SI) in a highly interconnected industrial environment. Rather than serving only quality control, metrology in Industry 4.0 now produces data flows that inform the whole manufacturing lifecycle.

2 The Foundations and Role of Metrology

Metrology serves as the foundation of measurement science and is the cornerstone of data reliability. European metrology is coordinated through collaboration between national metrology institutes (NMIs), and EU-level bodies, with the support of the European Commission. The central body facilitating this coordination and metrology research in Europe is the European Association of National Metrology Institutes (Euramet). The most influential National Metrology Institutes (NMIs) in Europe, such as the Physikalisch-Technische Bundesanstalt (PTB) in Germany and the National Physical Laboratory (NPL) in the United Kingdom, have developed digitalization strategies that guide the European direction. For example, a

pan-European consortium led by the PTB has initiated the development of a coordinated European Digital Quality Infrastructure (DQI) for new services and products, called the "European Metrology Cloud" (EMC). In this context, the term "cloud" refers to a distributed trusted metrology platform. Therefore, metrology at the European level is coordinated by Euramet for research and collaborative initiatives, with the PTB playing a strategic leadership role in the development of the digital infrastructure (EMC) in line with European Union regulations and global standards. Indeed, even at the global level, digital metrology is coordinated by bodies that directly influence European practice: this is the case with the International Committee for Weights and Measures (CIPM) that, in 2019, established a Digital-SI Task Group to develop a coordinated, uniform, authoritative, and reliable framework for data exchange based on the International System of Units (SI) (Thiel et al., 2017), (Euramet, M., 2022).

Traditional metrology focuses on accuracy and uncertainty assessment, aspects that remain essential even in the new scenario. National and international bodies guarantee metrological traceability according to shared criteria, ensuring the equivalence of calibration certificates worldwide. The breadth of metrological fields traditionally covered with the goal of maximum accuracy is well known: dimensional measurements (from nanometrology to long distances), thermometry (based on the International Temperature Scale ITS-90), mass and related quantities (force, pressure, hardness), and electromagnetic measurements (DC, AC, high voltage, radio frequency), the creation of the unit of time and frequency, etc..

3 Metrology Evolution Towards the 4th Revolution (Metrology 4.0)

The Fourth Industrial Revolution is a deep change, made possible by developments in metrology. Metrology now increasingly functions as a continuous production-level feedback mechanism rather than a separate after-the-fact inspection step.

3.1 New Requirements and Objectives

Industry 4.0 measurement systems must meet tight requirements — for instance, enabling calibration approaches that avoid production stoppages

Integration and Automation: Measurement must be seamlessly integrated into the production line and must aim for 100% product inspection, rather than just random sampling.

Real-Time Data: Measurement data must be provided in real time with the lowest possible uncertainty, in a format

that can be processed and transmitted throughout the entire process, from the sensor to the cloud.

Waste and Cost Reduction: The goal is to maximize efficiency, reduce downtime, and optimize maintenance.

3.2 Enabling Technologies (for Industry 4.0) and Metrology

Metrology 4.0 is intrinsically linked to the main technologies of Digital Transformation, which require real-time, secure, and traceable measurements. The evolution of metrology is crucial, as it provides the tools for the objective knowledge needed to make informed decisions and implement the transformation of a factory into a "smart factory". A wide range of technologies and tools are the pillars of this transformation.

3.2.1 Internet of Things (IoT) and Industrial IoT (IIoT)

IIoT adapts general IoT concepts for manufacturing: networks of sensors and devices collect streaming data for monitoring, optimization, and logistics planning. Metrology is central to IIoT systems. An example of this is the European SmartCom project, which is dedicated to defining the methodological and technical foundations for universal, secure, and unambiguous communication of metrological data in IoT and Industry 4.0 contexts (Barbosa et al., 2022), (Tagliaferri & Kuhl, 2024). Key elements are:

1. **Hardware Components and Edge Computing:** IIoT systems consist of an integrated network of devices that connect the enterprise ecosystem, including sensors, actuators, controllers, gateways, network switches, and industrial PCs (IPCs). These edge devices collect, process, and transmit data. Edge computing processes data close to its source to minimize latency and reduce network loads, which is critical for time-sensitive tasks like predictive maintenance.
2. **Communication and Protocols:** Specific connectivity protocols are used to ensure a continuous and secure flow of information between devices

3.2.2 Digital Twin (DT)

A DT is a dynamic virtual model of a physical asset or process; continuous measurement streams supply the data that update and validate twin behaviour. In other words, DTs use real-time measurements to mirror physical systems, enabling simulation and anomaly detection (Jiang et al., 2021), (Gohari et al., 2019). DT is a virtual model that faithfully represents a real product or process and is considered a cornerstone of Industry 4.0. Cyber-physical measurement systems and their data serve as an interface between the real production environment and its virtual representation. Real-time measurement data is the "material that digital twins are made of". DT also allows for performance monitoring, progress optimization, results simulation, and prediction of potential anomalies or failures. Using DT significantly reduces wasted time and resources.

3.2.3 Artificial Intelligence (AI) and Machine Learning (ML)

AI and ML are integrated to manage massive amounts of data and identify complex patterns. Applications include predictive analysis of instrument degradation, calibration plans optimization, and automated calibration. However, in metrology, deterministic algorithms and interpretable models (such as linear regression or decision trees) are preferred to ensure the repeatability, stability, and verifiability of results, which are essential for the validity of measurement processes. Self-learning models (such as deep learning) have fewer direct applications where maximum metrological transparency is required. AI and ML are essential for managing the large amounts of data (big data) generated in Metrology 4.0 (Wieczorowski & Trojanowska, 2021), (Varshney et al., 2022).

AI is used for predictive instrument analysis (to prevent deviations from specifications), calibration plan optimization, and automated calibration. AI can also support users in making decisions quickly and efficiently, leveraging data from the knowledge base (Wieczorowski & Trojanowska, 2021).

Despite the usefulness of machine learning, which allows systems to learn autonomously from data, metrology requires stability, repeatability, and verifiability. For this reason, deterministic algorithms and interpretable models (which produce understandable and verifiable results) are preferred over less transparent self-learning models such as deep learning.

3.2.4 Digitalization of Metrology Services and Safety

Digital transformation requires standardized tools and infrastructure to ensure data traceability and security. This includes defining minimum requirements for the exchange of machine-readable data in digital communication.

4 Strategic Areas - Focal Points

The evolution of metrology toward Industry 4.0 requires an internationally coordinated strategy to manage the integration of measurement into cyber-physical systems and the digital ecosystem (Sousa & Almeida, 2021).

The most influential approach to metrology digitalization is the strategy formulated by the PTB, the German NMI, in 2017, which is implemented through several projects focusing on four strategic focal points. (Barbosa et al., 2022), (Eichstädt, 2017).

These focal points represent the research and development agenda that many NMIs are addressing to create a Digital Quality Infrastructure (DQI) that ensures the traceability and reliability of measurements in the digital environment.

4.1 Focal Point 1 (FP1): Digital Transformation of Metrology Services

FP1 focuses on the digital upgrade of quality infrastructure and legal metrology. The digital transformation of metrology services creates value by extending previously unavailable capabilities, removing barriers to cross-border

e-commerce, and improving consumer protection (Gadelrab & Abouhogail, 2021), (Eichstädt, 2017), (Varshney et al. 2021). Key domains of FP1 include:

- Digital Calibration Certificates (DCCs): DCCs are intended to replace paper-based calibration certificates (PCCs) and require a secure digital infrastructure for authentication, validation, and complete traceability of the metrological chain back to primary samples (Mustapää et al., 2020), (Brown et al., 2020).
- Digital Representation of Metrological Traceability: Metrology 4.0 requires standardized, machine-readable formats to represent physical quantities, units of measurement, and, crucially, measurement results, including deviations and uncertainty components (Barbosa et al., 2022).
- Digitalization of the Quality and Legal Metrology Infrastructure: The goal is to develop reference architectures and establish a metrology cloud (such as the European Metrology Cloud - EMC) to support conformity assessment and market surveillance processes. The use of blockchain-based technologies is strongly emphasized for their ability to ensure integrity and traceability up to the highest level of the metrological chain (Oppermann et al., 2020), (Softicet et al., 2021).

4.2 Focal Point 2 (FP2): Metrology in Big Data Analysis

FP2 focuses on the development of metrological analytical methods to manage, archive, and ensure the security and reliability of large amounts of data (Big Data). Digital measurements generate a massive amount of data that must be transformed into strategic information (Gadelrab & Abouhogail, 2021). Key domains of FP2 include:

- Metrological Traceability in the IoT: It is essential to integrate metrological principles (SI traceability and uncertainty assessment) into the IoT data lifecycle, especially considering heterogeneous and rapidly evolving sensor networks.
- Use of AI, Machine Learning, Digital Twins, and Cloud Computing: Metrology in this context is central to predictive maintenance. AI and machine learning are used to analyze data, identify patterns, and optimize processes (such as calibration plans or automated calibration). Digital Twin relies on data from cyber-physical measurement systems as an interface between physical reality and virtual representation.
- Cyber-Physical Systems (CPS): CPS, often integrated with IoT and cloud computing to form cyber-physical manufacturing measurement systems (CPMM), aim to create factories with increasing intelligence and self-adaptation.

4.3 Focal Point 3 (FP3): Metrology of Communication Systems for Digitalization

This strategic point addresses the need to metrologically validate and ensure the security and efficiency of communication in complex industrial scenarios (Ačko et al., 2020). Key domains of FP3 include:

- Data-Driven Metrology Infrastructure: Requirements for machine-readable data exchange in digital communication must be defined. The European SmartCom project aims to provide methodological and technical foundations for universal, unambiguous, and secure communication of metrological data in IoT and Industry 4.0 networks.
- Integration of IoT, CPS, and Cloud Computing: These elements are designed to maximize efficiency and develop new services. The PTB, for example, has promoted the European Metrology Cloud (EMC) as a distributed, trusted platform to support compliance processes.
- Data Security and IIoT: Interconnection between devices increases vulnerabilities. Customized security solutions are needed for the IIoT value chain. The implementation of an inter-NMI blockchain network (such as the joint initiative between PTB and Inmetro in Brazil) is being addressed to connect NMIs globally and ensure the integrity and authenticity of legal information.

4.4 Focal Point 4 (FP4): Metrology for Simulations and Virtual Measuring Instruments

FP4 is dedicated to the development of analytical methods and licensing procedures for virtualized and interconnected measuring systems, as well as for the simulation of complex measurements for automated process control. The key domains of FP4 include:

- Virtual Metrology (VM) and Automatic Virtual Metrology (AVM): VM/AVM focuses on product quality estimation. Although originally developed for semiconductor manufacturing, it can be adapted to other industries adopting Industry 4.0. VM aims to achieve a faster and more reliable classification scheme for parts produced online and in real time (Chien et al., (2022).
- Deep Learning (DL) Integration: DL is used in virtual metrology contexts, especially for semi-automatic feature extraction and defect identification in large volumes of data.
- Augmented Reality (AR): AR is applied to automate and accelerate quality assurance tasks. For example, AR can assist operators in verifying component conformity in real time.
- The strategic areas do not simply digitize existing services but redefine the role of metrology, trans-

forming it into a dynamic, data-driven infrastructure that enables key Industry 4.0 technologies (such as AI, IoT, and Digital Twin). This strategy ensures that the fundamentals of metrology (accuracy and traceability) are maintained and promoted in a highly connected and automated manufacturing ecosystem.

5 Risks and Challenges

The integration of Metrology 4.0 with Industry 4.0 and the IIoT represents a revolution based on digitalization and real-time data analytics. While this offers enormous advantages in terms of efficiency, automation, and predictive maintenance, sources highlight a series of critical risks and challenges that must be addressed to ensure the credibility and reliability of digital measurements.

5.1 Risks Related to Cybersecurity and Data Integrity.

The greatest threat in the interconnected digital environment is data and infrastructure security, which is a top priority.

Cybersecurity as a Main Barrier: Digital transformation inevitably entails data security risks; therefore, it brings issues such as cybersecurity, data encryption, data storage, and secure communication. Stakeholders repeatedly identify cybersecurity, including data integrity and secure communication, as a principal barrier to IIoT deployment in metrology.

Increased Vulnerabilities: The interconnection between devices, machinery, and systems increases efficiency, but at the same time, exposes the infrastructure to potential attacks. Such attacks can manifest unauthorized access, operational disruptions, or data manipulation.

Data Manipulation: There is an inherent risk in maintaining the integrity and confidentiality of digitally stored data. Data manipulation or alteration is possible. Because measurements are critical, even a small error in a measurement can compromise product quality and cause financial losses.

Blockchain Reliability: Although blockchain technology is seen as a reliable solution for ensuring the integrity and authenticity of metrological data, its implementation must overcome challenges related to the large amount of data, privacy, and oracle authentication.

5.2 Metrology Challenges in Data Analytics (AI/ML)

The introduction of AI and ML involves a trade-off between the flexibility of big data analysis and the fundamental requirements of metrology.

Stability and Repeatability: The adoption of AI must address the challenge of ensuring stability, repeatability, and verifiability of results, which are crucial for the validity of measurement processes.

Non-Interpretable Models: The most complex and self-learning models, such as those based on DL and Natural Language Processing (NLP), while powerful, do not offer the stability and repeatability required for metrology. Their results can vary from one run to the next, making it difficult to document a standard metrology process. For

this reason, metrology tends to favor deterministic algorithms and interpretable models (such as Linear Regression or Decision Trees).

Big Data Challenges: The challenges for data analysis in a data-driven industrial approach are classified into four main groups: data size, the choice of DL models, the nature of the data, and the uncertainty associated with DL models generated at laboratory scale.

Digital Representation of Uncertainty: Digitalization requires a new approach to the digital representation of measurement uncertainty. Reporting formats should include uncertainty components, which are not in line with current practices, but are necessary for digital systems to use this information in end-use applications.

5.3 Technical and Interoperability Challenges

The implementation of IIoT and Metrology 4.0 in a heterogeneous industrial context raises issues of integration and the technical performance of sensors.

Interoperability and Integration Costs: Industrial infrastructures often use devices and systems from different manufacturers, each with different standards, languages, and protocols. This lack of uniform compatibility can hinder the full functionality of IIoT systems and increase integration costs.

Unreliable Connectivity: All IoT devices and equipment require a stable and reliable connection to function properly. Service interruptions can compromise operational efficiency. In the IIoT scenario, the inherent reliability and predictability of wireless connections (used in mobile or wide-area scenarios) remain extremely challenging.

Limitations of Measurement Methods: Metrology 4.0 systems impose stringent requirements. For example, optical measurements (which generate large amounts of data) require almost perfect surface cleanliness and are sensitive to external factors. Conversely, tactile metrology is slow due to the need for mechanical contact and can damage sensitive objects.

Privacy and Big Data Management: IIoT systems generate enormous volumes of data.

5.4 Organizational and Strategic Challenges

The adoption of digital transformation is not just a technological issue, but also a strategic and organizational one. **High Failure Rates:** Despite the numerous benefits, some research reports a high failure rate associated with the implementation of digital transformation.

Need for Strategy: The success of digitalization depends on its strategic adoption within the organization.

Trust and Transparency: It is essential to ensure transparency in the use of collected information and define clear responsibilities to ensure ethical use and earn the trust of companies.

6 Benefits and Outlook

Metrology 4.0, integrating accurate and reliable measurements with real-time data analysis, is radically transforming manufacturing processes. Benefits include productivity gains and lower operating costs: automated calibration

scheduling and continuous monitoring permit interventions only when indicators suggest deviation.

Improved quality: the implementation of intelligent and adaptive quality strategies ensures batch consistency and reduces raw material waste.

Collaborative innovation: Metrology 4.0 enables innovative and personalized actions thanks to increasingly collaborative conclusions in the decision-making process.

7 Conclusions

Metrology 4.0 represents a technological revolution powered by Industry 4.0. It is dynamically and automatically integrated into the manufacturing process. The primary objective is to provide real-time data for continuous quality control and optimization. This evolution is constant, driven by research into new technologies, methods, and techniques aimed at improving various aspects, such as reducing costs, increasing quality, and reducing waste. Looking to the future, the evolution towards Industry 5.0 promises further improvements, placing greater emphasis on human-machine collaboration to ensure even higher levels of traceability, repeatability, and reproducibility. In conclusion, the digitalization of metrology is serving as an essential knowledge input for the production system, assuming a crucial role in decision-making processes and ensuring the success of production lines. Metrology 4.0 integrates measurement data into decision workflows and acts as a bridge between physical production and its virtual representations, enabling data-driven optimization.

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